3DBionotes COVID-19 Structural Hub:

A Central Resource for Validation Information and Refined Models

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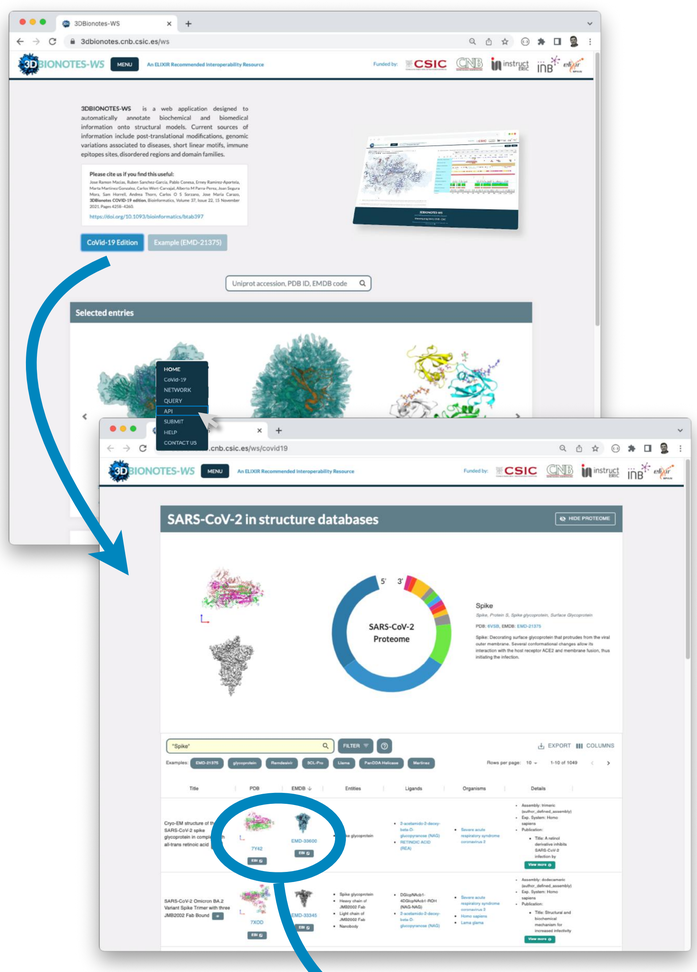
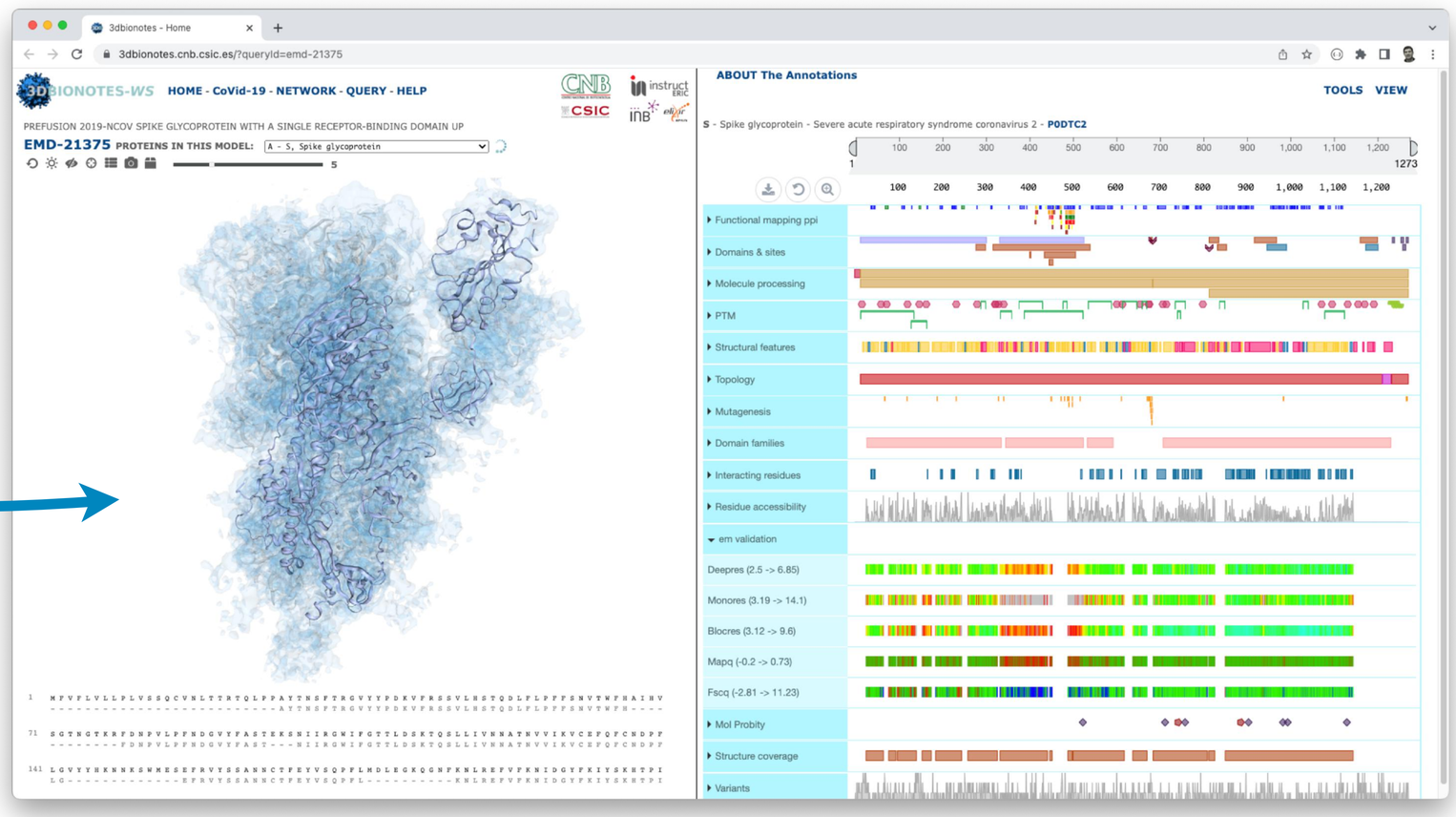


3DBIONOTES-WS

3DBionotes-WS is a set of Web Services providing multiple annotations oriented to structural biology analysis, plus a website interface including a 3D viewer for macromolecular models and Cryo-EM derived maps. The display is fully interactive with a corresponding display for genomic, proteomic and structural feature annotations. With the COVID-19 outbreak, easy access to quality data for the required structural analysis of SARS-CoV-2 and other related viruses and its interactions with human macromolecules become a necessity. We have set up a new section in the website aiming to collect and provide all available structural information available.

Apart from some interesting examples, we have added a simplified schema of the proteome of the virus that can be used as an index, with links to the corresponding section for every protein of the virus. There, we have a panel with a brief description and its role in the virus life cycle. the search box and filters allow quick access to any entry of our interest within the database.

The most relevant information for each entry is displayed in the different columns, which can be rearranged as needed and allows us to sort the results by any of the columns. The images here, are links to show that corresponding entry in our 3D Viewer.



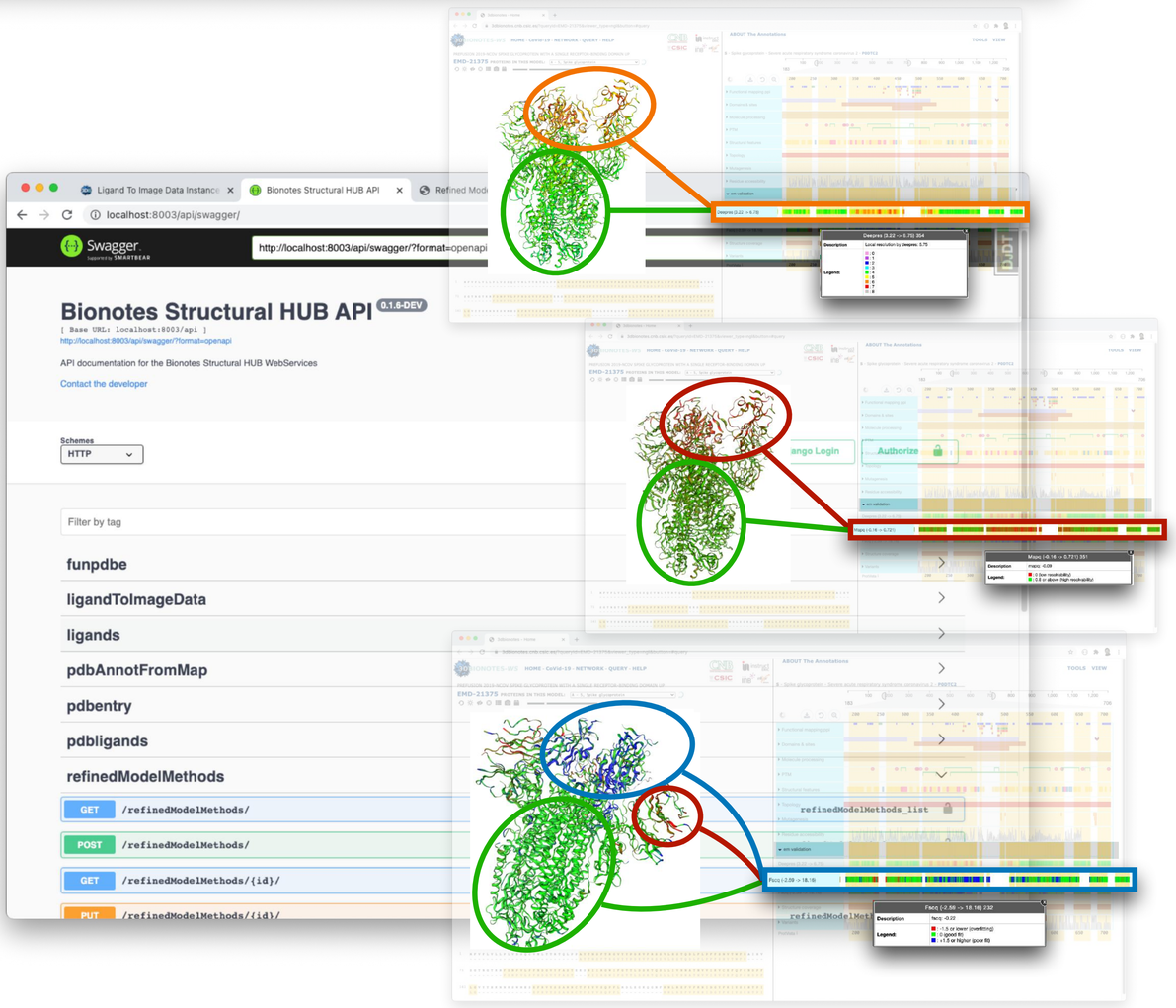
CRYO-EM VALIDATIONS

In line with previous concerns raised, our group has aligned with other initiatives to bring more attention not only to the quantity of data, but also to the quality of the data.

With the COVID-19 Structural Hub we have dedicated a website section to collect and provide easy access to all available structural information related to the topic. New annotation types have recently been added with validation information about cryo-EM maps, atomic models, and map-to-model fitting. In addition to re-refined models by PDB-REDO, CERES and the Coronavirus Structural Task Force, we provide annotations for local resolution validation methods like DeepRes, MonoRes and BlocRes. If an atomic model is available, we also compute MapQ and FSC-Q scores, a quantitative estimation of how much of the model is supported by the signal content of the map.

This way is quite easy to find interesting regions of the structure with similar resolution values and get a better understanding of the flexibility and conformational dynamics of a structure.

All these new annotations are now part of our reach Application Programming Interface (API), that can be easily integrated into further bioinformatic tools. Full documentation is provided using the Swagger UI for convenience, including tools for testing and sending example queries on-line.



https://3dbionotes.cnb.csic.es/ws/api

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